Health and Technology (2026) 16:547–554
https://doi.org/10.1007/s12553-026-01068-9

ORIGINAL PAPER

Explainable and accurate ENT diagnosis via a dual knowledge LLM

Jing Gong[1] · **Hao Yuan**[2] · **Fuchun Li**[2]

Received: 11 November 2025 / Accepted: 26 March 2026 / Published online: 6 April 2026
© The Author(s) under exclusive licence to International Union for Physical and Engineering Sciences in Medicine (IUPESM) 2026

Abstract

Purpose Artificial intelligence (AI), especially large language models (LLMs), faces a clinical trust gap due to black-box reasoning and factual hallucinations–risks amplified in otorhinolaryngology (ENT), where diagnostic pathways are complex and symptoms overlap. We aim to build an explainable and accurate ENT diagnostic assistant grounded in a dual-knowledge framework.

Methods We construct a rule-centric *Core Knowledge Graph* (KG) distilled from peer-reviewed guidelines and handbooks, and a *Contextual Instance Library* (CIL) of real-world case narratives. At inference, the system performs dual retrieval–logical KG reasoning for citable rules and dense similarity search for experiential corroboration–then applies schema-constrained prompting to synthesize both streams into auditable reports. Offline, KG/CIL are automated with LLM-based NER/RE, discourse-aware chunking, and vector indexing. Evaluation uses a 251-item ENT question set and curated vertigo cases, compared against base LLMs and standard RAG baselines; explanation quality is assessed with LLM-as-a-judge across multiple generator/judge pairs.

Results The dual system achieves higher diagnostic accuracy than base LLMs and standard RAG baselines and produces consistently superior, citation-backed explanations. Reports explicitly link conclusions to rules and corroborating instances.

Conclusions Integrating rule knowledge (KG) with experiential evidence (CIL) simultaneously improves diagnostic accuracy and interpretability, offering a pragmatic path from black-box LLMs to trustworthy, auditable clinical decision support in ENT.

Keywords Retrieval-augmented generation · Knowledge graph · Explainability · Otorhinolaryngology · Vertigo · BPPV · Ménière · Clinical decision support

1 Introduction

Recent advances in large language models (LLMs) promise to transform healthcare by processing vast unstructured clinical text (notes, reports, literature). Yet two barriers hinder adoption in high-stakes diagnosis: (i) *accuracy and reliability*, as base LLMs may hallucinate plausible but false statements when facing specialized and rapidly evolving medical knowledge; and (ii) *explainability and trust*, since opaque reasoning impedes auditability and accountability in clinical workflows [1, 2].

These barriers are acute in otorhinolaryngology (ENT) differential diagnosis, where overlapping symptoms (e.g., vertigo, tinnitus, hearing loss) require guideline-grounded reasoning and attention to red flags. For vertigo, the Dix–Hallpike test is pivotal for benign paroxysmal positional vertigo (BPPV) [7], while the HINTS battery helps distinguish central strokes from peripheral vestibulopathies in acute vestibular syndrome [8]. In practice, clinicians combine *rule knowledge* (guideline-derived diagnostic relations) with *experiential knowledge* (patterns learned from prior cases). Current AI approaches often emphasize one side: vector-only RAG captures experience but lacks

✉ Fuchun Li
lifuchun25@ime.ac.cn

Jing Gong
gongjingzgc@foxmail.com

Hao Yuan
yuanhao241@mails.ucas.ac.cn

[1] Beijing Zhongguancun Hospital, No. 12 Zhongguancun South Road, Beijing, China

[2] Chinese Academy of Sciences, No. 52 Sanlihe Road, Beijing, China

explicit diagnostic logic, whereas KG-only reasoning offers interpretability but can be brittle in coverage [3].

We hypothesize that *explicitly separating and then integrating* rule knowledge and experiential knowledge yields both accuracy and auditability. We therefore propose a dual knowledge system comprising a rule-centric Core Knowledge Graph distilled from peer-reviewed guidelines and a Contextual Instance Library of de-identified case narratives; at inference, the system retrieves complementary rule and case evidence and generates a structured, source-attributed diagnostic report.

Figure 1 contrasts the opacity of base LLM outputs with our end-to-end auditable report that cites both KG rules and CIL cases.

Our contributions are:

- **Modeling complex ENT differentials.** We operationalize expert-like reasoning in ENT by jointly leveraging rules (guidelines) and experience (cases).
- **Dual knowledge architecture.** We introduce a Core KG with explicit diagnostic relations (*confirms*, *rules_out*, *manifests_as*, *treated_by*, *differentiates_from*) complementary to a vectorized CIL [4–6].
- **Training-free construction and evaluation.** We automate KG/CIL construction and quantify explanation quality via LLM-as-a-Judge across multiple generator/judge pairs [10, 11].

2 Related work

2.1 Large language models for medical diagnosis

General-purpose and biomedical LLMs have rapidly advanced from domain-pretrained encoders (e.g., PubMedBERT) to generative models tailored to biomedical text (e.g., BioGPT) and very large clinical models trained on EHR corpora (e.g., GatorTron), showing strong results on MedQA, PubMedQA and related benchmarks [1, 10–14]. Nevertheless, the gap between exam-style QA and bedside decision support remains substantial: surveys and controlled studies highlight hallucination, overconfident misstatements, and limited calibration/uncertainty reporting in high-stakes clinical settings [15, 16]. Retrieval-augmented generation (RAG) addresses factuality by conditioning generation on retrieved evidence; along this line, REALM

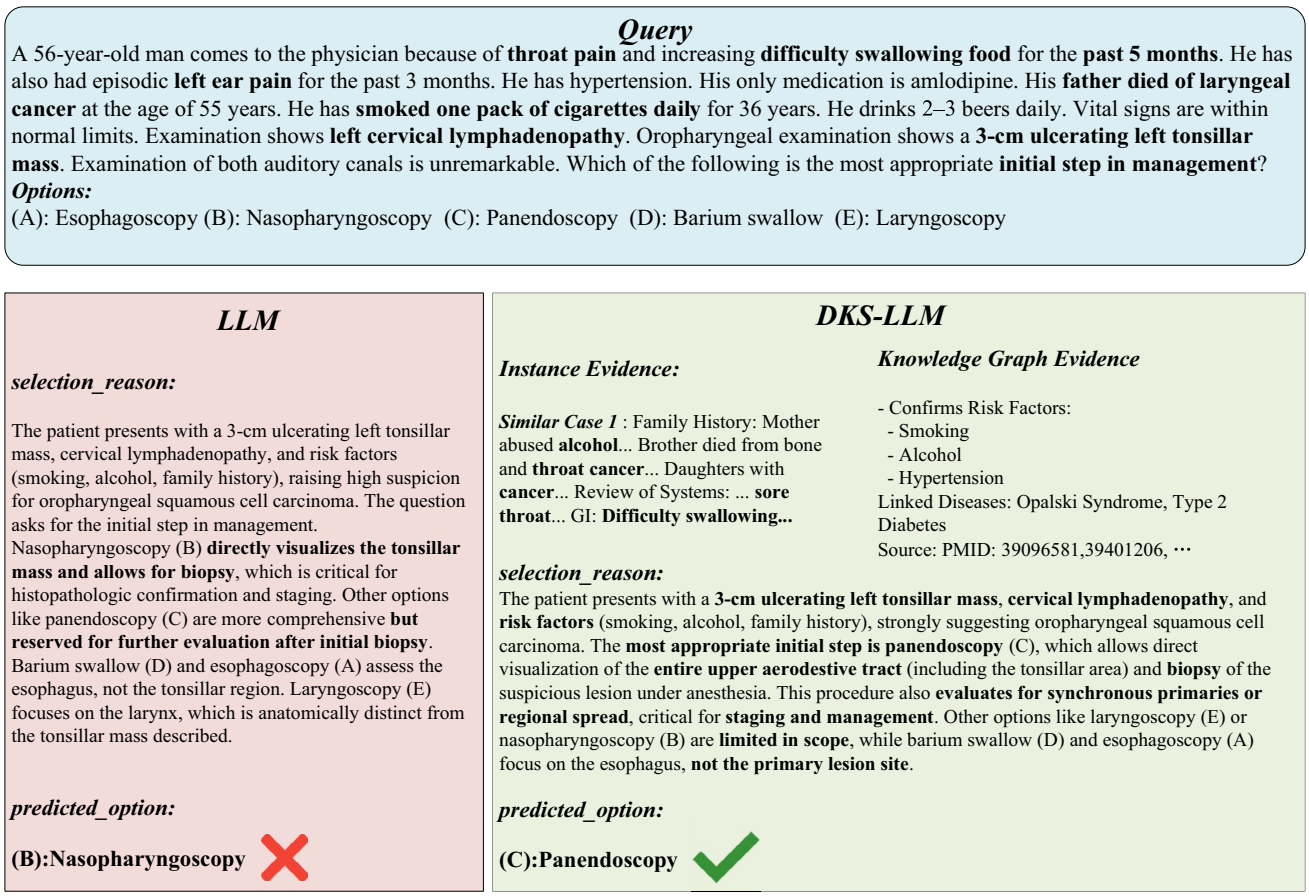


Fig. 1 Comparison between a standard LLM output (black-box) and our dual knowledge system output (rule paths + case evidence with citations)

incorporated differentiable retrieval during pretraining [17], DPR boosted dense passage retrieval quality for open-domain QA [18], FiD aggregated multiple retrieved passages through a seq2seq decoder [19], RETRO scaled parametric-complemented retrieval to trillion-token corpora [20], and standard RAG formulations have become strong baselines in knowledge-intensive NLP [2]. Yet, purely vector-based RAG can struggle in clinical differential diagnosis where reasoning requires explicit assertion types (e.g., confirmation vs. exclusion), disease–symptom temporal logic (onset order, triggers), and the ability to distinguish the evidential weight of heterogeneous sources (guidelines > reviews > single case reports). In addition, sparse retrieval (e.g., BM25) and dense semantic retrieval (e.g., Sentence-BERT) capture complementary signals–lexical precision versus semantic similarity–but both are sensitive to discourse segmentation of clinical narratives, motivating discourse-preserving chunking and scalable nearest-neighbor search (e.g., FAISS) for case-based evidence [9, 24, 25]. Our work situates itself in this landscape by preserving rich narrative structure in a Contextual Instance Library while constraining generation to cite retrieved passages, and by pairing instance evidence with symbolic, guideline-grounded reasoning to reduce unsupported claims and improve auditability beyond vanilla RAG.

2.2 Medical knowledge graphs and explainability

Medical knowledge graphs (KGs) and ontologies such as UMLS and SNOMED CT provide curated, typed biomedical relations and concept normalizations that enable auditable, symbolic reasoning over entities and edges [21, 22]. Recent surveys on GraphRAG suggest that fusing graph structure with neural retrieval can improve transparency, compositional reasoning, and provenance control for knowledge-intensive tasks [3]. However, KG-only pipelines are often brittle in coverage (schema incompleteness, concept drift) and may be overly doctrinal when real-world cases present atypical trajectories; conversely, vector-only systems lack hard relational semantics (e.g., *confirms*, *rules_out*, *manifests_as*, *treated_by*, *differentiates_from*) that clinicians use in differential diagnosis. Prior work on biomedical IR and QA (e.g., BioASQ) has emphasized the importance of gold evidence and justification, while the XAI literature (e.g., LIME/SHAP) cautions that post-hoc rationales may not reflect the actual decision pathway [23, 26, 27]. Complementarily, "LLM-as-a-judge" protocols provide scalable comparative assessment for long-form answers, but require careful design to avoid self-judging biases and to align criteria with domain-specific expectations [28]. Our contribution operationalizes a division of labor where a Core KG, instantiated from guideline/handbook sources, drives graph-first hypothesis discrimination with per-edge provenance, while a case-based retriever supplies experiential corroboration from clinically faithful narratives; this dual-knowledge synthesis yields guideline-consistent, source-attributed explanations and higher diagnostic accuracy than KG-only or vector-only baselines, thereby addressing both faithfulness and interpretability gaps identified in prior work [2, 3, 17–22].

3 Methodology

3.1 Overall architecture

We design a multi-stage diagnostic pipeline that mirrors expert clinical reasoning, consisting of query understanding, parallel evidence retrieval, and structured synthesis. Figure 2 provides an overview of the system.

Query understanding Given a free-text clinical case, an LLM-based extractor identifies medically salient entities spanning *Symptom*, *Disease*, *Test*, and *Treatment*. These entities form an explicit and interpretable representation of the input that anchors subsequent retrieval and reasoning. For example:

> *Input:* "45-year-old male with recurrent episodes of vertigo lasting hours, associated with tinnitus and right-sided hearing loss."
>
> *Output:* "vertigo (Symptom), tinnitus (Symptom), hearing loss (Symptom)."

Dual retrieval Two complementary retrieval processes operate in parallel. (*i*) KG-based reasoning (rule evidence). From the extracted entity set $E_Q = \{e_1, \ldots, e_k\}$, we traverse the Core Knowledge Graph to identify candidate diseases that are diagnostically connected to multiple query entities via typed relations. Each candidate disease d is scored by the degree of evidential convergence:

$$S_{\mathrm{KG}}(d) = \sum_{e \in E_Q} \mathbb{I}\big((d, r, e) \in G \vee (e, r, d) \in G\big),$$

where r ranges over clinically meaningful relations such as *confirms*, *rules_out*, *manifests_as*, *treated_by*, and *differentiates_from*. Higher scores indicate stronger rule-based diagnostic support. (*ii*) Instance retrieval (experiential evidence). In parallel, the full case narrative is encoded into a dense vector v_q to preserve temporal and discourse cues. Similarity search over the Contextual Instance Library yields the top-K most relevant case fragments based on cosine similarity:

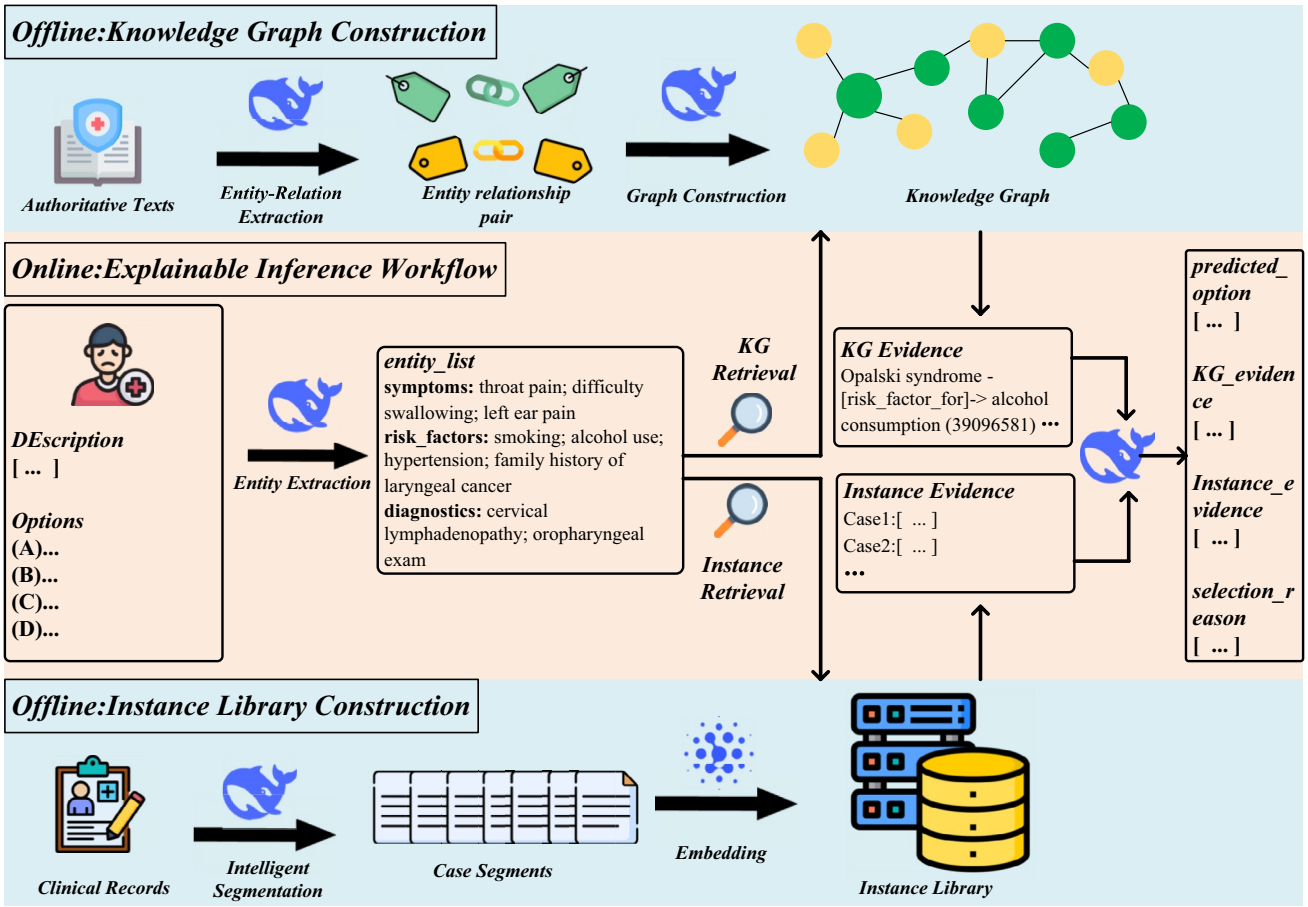


Fig. 2 System architecture: dual retrieval from the Core Knowledge Graph (KG) and Contextual Instance Library (CIL), followed by structured generation of an auditable diagnostic report

$$S_{\mathrm{CIL}}(Q,c) = \frac{\mathrm{v}_q \cdot \mathrm{v}_c}{\|\mathrm{v}_q\| \, \|\mathrm{v}_c\|}.$$

Structured generation Retrieved rule paths and case snippets are integrated by a schema-constrained generation prompt that produces a diagnostic report comprising: (i) candidate diagnoses; (ii) supporting rule evidence with provenance; (iii) corroborating case evidence; and (iv) safety considerations and red flags. The generator is explicitly restricted to using retrieved evidence, reducing unsupported claims.

3.2 Explainability by design

Explainability is treated as an intrinsic system property rather than a post-hoc textual justification. It is enforced through architectural constraints at each stage of the pipeline.

Explicit grounding Entity-based query understanding ensures that downstream reasoning operates on clinically interpretable units rather than latent prompt representations.

Typed rule reasoning Diagnostic reasoning over the Core KG relies on typed relations aligned with clinical semantics (e.g., confirmation vs. exclusion), and supporting graph paths are explicitly exposed to enable auditability.

Dual-source provenance Explanations are grounded in two complementary evidence sources–guideline-derived rules (KG) and experiential cases (CIL)–each accompanied by explicit provenance, allowing clinicians to distinguish normative guidance from case-based corroboration.

Structured outputs The fixed report schema requires that every diagnostic claim be supported by retrieved evidence, transforming explanations from free-form narratives into auditable reasoning artifacts.

3.3 Core knowledge graph (rule knowledge)

Sources and schema The Core Knowledge Graph encodes entities (*Disease*, *Symptom*, *Test*, *Treatment*) and diagnostic relations tailored for ENT differentials, including *confirms*, *rules_out*, *manifests_as*, *treated_by*, and *differentiates_from*. Graph content is distilled from peer-reviewed guidelines and consensus criteria in vertigo and related ENT conditions, such as BPPV and Ménière's disease updates, as well as vestibular migraine criteria.

Extraction and traceability LLM-based named entity recognition and relation extraction operate on guideline paragraphs. Each extracted triple retains bibliographic metadata and quote spans to preserve provenance. At inference time, only structured triples and path snippets are exposed, ensuring auditable rule-based reasoning without verbatim text reproduction.

3.4 Contextual instance library (experiential knowledge)

Agentic chunking Clinical narratives are segmented by an LLM-based curator into discourse-consistent units (e.g., chief complaint, history of present illness, examination, assessment) to preserve semantic coherence during retrieval.

Vectorization and indexing Each chunk is embedded into a 1024-dimensional vector and indexed using FAISS for efficient approximate nearest neighbor search. Retrieved chunks are linked back to their original sources via stable identifiers, enabling verification and contextual inspection.

4 Experiments

4.1 Datasets and task

We evaluate the proposed system on two complementary benchmarks that reflect both objective diagnostic accuracy and clinically realistic reasoning.

ENT-MedQA-251 ENT-MedQA-251 is a curated subset of MedQA-USMLE containing 251 multiple-choice questions focused on otorhinolaryngology scenarios, including history, physical examination, diagnostic workup, and management. Each item includes a clinical vignette, four or five candidate options, and a single gold answer [10]. We remove duplicated stems, exclude items with ambiguous keys, and normalize option text to ensure consistent evaluation. Performance on this dataset is measured using Accuracy@1.

Vertigo-focused case set The second benchmark consists of narrative-style clinical cases covering benign paroxysmal positional vertigo (BPPV), Ménière's disease, vestibular migraine, vestibular neuritis, and clinically relevant central causes. Labels and red-flag criteria are derived from AAO-HNS guidelines and Bárány Society consensus updates [4–6]. This dataset emphasizes temporal progression, symptom co-occurrence, and differential reasoning beyond multiple-choice selection.

Across both benchmarks, the task is to generate a structured diagnostic report including a final diagnosis, a differential with justification, and recommended next steps; for ENT-MedQA-251, the system must additionally select the correct option.

4.2 Dataset usage and evaluation workflow

The proposed dual knowledge system operates in a strictly *training-free* manner with respect to all evaluation datasets. No backbone LLM parameters are updated using either ENT-MedQA-251 or the vertigo-focused case set.

Workflow separation The experimental workflow is explicitly divided into three phases: (*i*) Offline knowledge construction, in which peer-reviewed guidelines and consensus documents are processed to build the Core Knowledge Graph (KG), and de-identified clinical narratives are curated into the Contextual Instance Library (CIL); (*ii*) Online inference, in which test queries undergo entity extraction, dual retrieval from the KG and CIL, and schema-constrained generation using a frozen backbone LLM; and (*iii*) Evaluation, in which generated outputs are assessed using accuracy metrics and LLM-as-a-Judge protocols. At no stage are evaluation items used to update model parameters.

Leakage control To prevent information leakage and circular reasoning, guideline texts are distilled into typed graph triples with bibliographic provenance, and verbatim sentences are never exposed at inference time. Evaluation questions are excluded from KG and CIL construction. During inference, the generator can access only retrieved KG paths and case snippets, each explicitly cited, ensuring that conclusions are grounded in auditable evidence rather than memorized text.

4.3 Baselines and variants

We compare four retrieval and generation configurations across multiple LLM backbones (*qwen*, *r1*, *v3.2*) to isolate the contribution of rule knowledge and experiential evidence:

- ***llm_only***: zero-shot generation without external evidence.
- ***vec_only***: standard retrieval-augmented generation using top-K case chunks from the CIL [2].
- ***kg_only***: rule-based reasoning using only KG paths and citations.
- ***full***: the proposed dual system combining KG reasoning with CIL retrieval.

Decoding parameters are fixed per backbone, and all runs use three random seeds.

4.4 Metrics and implementation

Accuracy We report Accuracy@1 on ENT-MedQA-251. For the vertigo-focused case set, we evaluate exact match

Table 1 Diagnostic accuracy on the ENT set (higher is better)

Backbone (LLM)	full (ours)	vec_only	kg_only	llm_only
qwen	0.880	0.857	0.861	0.841
r1	0.884	0.884	0.873	0.865
v3.2	0.817	0.817	0.813	0.809
gpt4o	0.865	0.853	0.849	0.837

against reference diagnoses mapped to a controlled vocabulary.

Explainability metrics We assess explanation quality using evidence coverage, rule–experience balance, and contradiction rate, reflecting the degree to which diagnostic claims are supported by retrieved evidence.

LLM-as-a-Judge A blinded ranking protocol compares reports from different system variants along logic clarity, evidence relevance, clinical reasoning fidelity, and overall trustworthiness [28].

Implementation details CIL chunks are embedded into 1024-dimensional vectors and indexed using FAISS [9]. KG triples retain bibliographic provenance, and inference-time prompts enforce a fixed schema requiring explicit evidence citation. All experiments are conducted on identical hardware, and latency is reported as median end-to-end inference time.

5 Results

5.1 Diagnostic accuracy

Table 1 reports Accuracy@1 across different LLM backbones and retrieval configurations. Across all backbones, retrieval-augmented variants consistently outperform the zero-shot *llm_only* baseline, confirming that ENT diagnosis benefits substantially from explicit grounding in external knowledge rather than relying solely on parametric memory.

The proposed dual system (*full*) achieves the best or tied-best performance for all evaluated backbones. On *qwen*, *full* attains an accuracy of 0.880, exceeding both *vec_only* (0.857) and *kg_only* (0.861). This gap suggests that experience-only retrieval can be susceptible to semantically similar but diagnostically irrelevant cases, while rule-based constraints from the Core Knowledge Graph help regularize decisions and suppress spurious matches. On *r1* and *v3.2*, *full* matches the strongest alternative, indicating that the added explainability afforded by KG grounding does not compromise predictive accuracy.

Notably, the same trend holds for the stronger backbone *gpt4o*. While *gpt4o* already demonstrates high baseline performance in the *llm_only* setting (0.837), incorporating retrieval improves accuracy across all variants. The dual system further raises performance to 0.865, outperforming both *vec_only* (0.853) and *kg_only* (0.849). This result indicates that even for advanced LLMs with strong internal knowledge, explicit integration of rule-based and experiential evidence provides complementary benefits rather than diminishing returns.

Overall, these findings support our central hypothesis that explicitly combining guideline-derived rule knowledge with case-based experiential retrieval yields a robust and backbone-agnostic pathway to high diagnostic accuracy in ENT scenarios.

Table 2 LLM-as-a-Judge: average rank (lower is better) across generator/judge combinations (excluding self-judging)

Generator	Judge	full	vec_only	kg_only	llm_only
qwen	r1	1.088	2.092	2.964	3.857
	v3.2	1.108	2.131	2.984	3.777
	gpt4o	1.359	2.044	2.928	3.669
r1	qwen	1.167	2.351	2.896	3.586
	v3.2	1.088	2.092	2.964	3.857
	gpt4o	1.323	2.084	2.833	3.761
v3.2	qwen	1.255	2.331	2.777	3.637
	r1	1.462	2.315	2.849	3.375
	gpt4o	1.466	2.080	2.825	3.629
gpt4o	qwen	2.191	2.438	2.721	2.649
	r1	1.849	2.498	2.538	3.116
	v3.2	1.367	2.291	2.845	3.498
Average	All judges	1.394	2.229	2.844	3.534

5.2 Explainability (LLM-as-a-Judge)

Table 2 reports blind-ranking results across multiple generator–judge combinations; the *full* system consistently achieves the lowest (best) average ranks, with a global mean rank of 1.394 when all generators and judges, including *gpt4o*, are taken into account, and the relative ordering among methods remains stable across heterogeneous generator and judge models.

A clear and consistent ordering emerges across almost all configurations: *full* $\gg$ *vec_only* > *kg_only* $\gg$ *llm_only*. While vector-only retrieval improves explanation quality over zero-shot generation by grounding responses in retrieved cases, it often lacks explicit logical structure. Conversely, KG-only reasoning provides clear rule-based justifications but may appear rigid or incomplete when confronted with atypical presentations. By integrating both evidence streams, the dual system produces explanations that are simultaneously structured, contextually rich, and clinically plausible.

Importantly, this pattern persists when *gpt4o* is used as either generator or judge. Even when strong models evaluate one another, explanations generated by the dual system are consistently preferred, suggesting that the improvements stem from the architecture and evidence integration strategy

rather than stylistic artifacts. These results demonstrate that the proposed approach enhances interpretability in a model-agnostic manner, aligning generated rationales more closely with expert clinical reasoning.

5.3 Qualitative analysis: case study

On a representative head-and-neck case (56-year-old male with five months of sore throat and dysphagia, three months of ipsilateral otalgia, left cervical lymphadenopathy, and a 3 cm ulcerative tonsillar mass; 36 pack-years, daily alcohol, family history of laryngeal cancer), the dual system recommends panendoscopy as the initial comprehensive evaluation. The KG contributes auditable rule paths that encode synergistic risk of tobacco and alcohol for head and neck malignancy and guideline-consistent workup emphasizing mucosal survey and biopsy for suspicious oropharyngeal lesions, while CIL retrieval surfaces patterns linking concurrent throat symptoms with family cancer history, clarifies the lack of systemic infection signs, and contrasts benign chronic ear complaints with *new-onset* ipsilateral otalgia suggestive of referred pain from an oropharyngeal primary. The structured report thus integrates rule constraints and experiential corroboration into a transparent rationale that mirrors expert practice and is fully traceable to sources.

6 Conclusion

This work tackles two persistent barriers for deploying LLMs in high-stakes otorhinolaryngology (ENT) diagnosis–accuracy and explainability–by introducing a dual knowledge system that functionally separates and then integrates rule knowledge distilled into a Core Knowledge Graph (KG) and experiential knowledge encoded in a Contextual Instance Library (CIL). At inference time, convergence-based KG reasoning provides logically discriminative, auditable paths among look-alike presentations (e.g., BPPV, Ménière's disease, vestibular neuritis, vestibular migraine), while nearest-neighbor retrieval over full narrative cases refines decisions with pattern-level corroboration; a schema-constrained generator fuses both streams to produce source-attributed reports that clinicians can audit. Across a 251-item ENT set and curated vertigo cases, the system achieves state-of-the-art or parity diagnostic accuracy against strong baselines and consistently superior explanation quality under LLM-as-a-judge assessments, demonstrating that explicit rule grounding plus experience-informed refinement is more reliable than vector-only RAG or KG-only pipelines [2, 3, 9]. Clinically, making guideline-anchored reasoning explicit (e.g., Dix–Hallpike maneuver for BPPV and red-flag triage with HINTS for central causes) strengthens bedside safety and aligns the AI's rationale with expert practice [4–8]. Limitations remain: KG coverage mirrors curated literature and revision cycles; CIL composition may bias retrieval toward atypical yet semantically close narratives; and residual hallucination risk stems from the backbone model despite schema constraints. Mitigations include entity-first parsing to seed precise KG queries, strict citation enforcement so unsupported claims are flagged or dropped, periodic KG/CIL refresh tied to peer-reviewed updates (rather than web sources), and safety overlays for red flags in acute vestibular presentations. Beyond ENT, the training-free, portable blueprint generalizes to specialties with strong guideline ecosystems and rich narrative data; prospective clinician-in-the-loop studies, improved uncertainty calibration and selective abstention, and multilingual deployments that preserve citation fidelity are promising next steps toward transforming LLMs from opaque tools into trustworthy, auditable clinical partners [1].

Author Contributions Jing Gong led the work, developed the methodology, conducted the experiments and data analysis, and drafted the manuscript. Hao Yuan collected and organized the relevant materials and data. Fuchun Li provided critical feedback, helped shape the analysis and interpretation, and revised the manuscript. All authors read and approved the final manuscript.

Funding This work was supported by the Research Project of Beijing Zhongguancun Hospital (Grant No. Y202413).

Declarations

Competing Interests The authors have no relevant financial or non-financial interests to disclose.

Ethics Approval Not applicable.

Consent to Participate Not applicable.

Consent to Publish Not applicable.

References

1. Singhal K, Tu T, Gottweis J, et al. Toward expert-level medical question answering with large language models. Nat Med. 2025;31(3):943–50. https://doi.org/10.1038/s41591-024-03423-7.
2. Lewis P, Perez E, Piktus A, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. In: NeurIPS. 2020.
3. Peng B, Zhu Y, Liu Y, et al. Graph Retrieval-Augmented Generation: A Survey. arXiv. 2024. https://arxiv.org/abs/2408.08921
4. Bhattacharyya N, Gubbels SP, Schwartz SR, et al. Clinical practice guideline: Benign paroxysmal positional vertigo (Update). Otolaryngol-Head Neck Surg. 2017;156(3 Suppl):S1–47. https://doi.org/10.1177/0194599816689667.
5. Basura GJ, Adams ME, Monfared A, et al. Clinical practice guideline: Ménière's disease. Otolaryngol-Head Neck Surg.

2020;162(2 Suppl):S1–55. https://doi.org/10.1177/0194599820909438.
6. Lempert T, Olesen J, Furman J, et al. Vestibular migraine: Diagnostic criteria (Update). J Vestib Res. 2022;32(1):1–6. https://doi.org/10.3233/VES-201644.
7. Dix MR, Hallpike CS. The pathology, symptomatology and diagnosis of certain common disorders of the vestibular system. Proc R Soc Med. 1952;45(6):341–54.
8. Kattah JC, Talkad AV, Wang DZ, Hsieh YH, Newman-Toker DE. HINTS to diagnose stroke in the acute vestibular syndrome: three-step bedside oculomotor examination more sensitive than early MRI. Stroke. 2009;40(11):3504–10. https://doi.org/10.1161/STROKEAHA.109.551234.
9. Johnson J, Douze M, Jégou H. Billion-scale similarity search with GPUs. arXiv:1702.08734. 2017.
10. Jin D, Pan E, Oufattole N, et al. What Disease does this Patient Have? A Large-scale Open Domain Question Answering Dataset from Medical Exams. 2020. https://doi.org/10.48550/arXiv.2009.13081
11. Jin Q, Dhingra B, Liu Z, Cohen WW, Lu X. PubMedQA: A dataset for biomedical research question answering. In: EMNLP-IJCNLP; 2019. p. 2567–2577.
12. Gu Y, Tinn R, Cheng H, et al. Domain-specific language model pretraining for biomedical natural language processing. ACM Trans Comput Healthc. 2021;3(1):1–23.
13. Luo R, Sun L, Xia Y, Qin T, Zhang S, Poon H, Liu T-Y. BioGPT: generative pre-trained transformer for biomedical text generation and mining. Brief Bioinform. 2022;23(6):bbac409.
14. Yang X, Chen A, PourNejatian N, et al. A large language model for electronic health records. npj Digit Med. 2022;5:194.
15. Ji Z, Lee N, Frieske R, et al. Survey of hallucination in natural language generation. ACM Comput Surv. 2023;55(12):248.
16. Maynez J, Narayan S, Bohnet B, McDonald R. On faithfulness and factuality in abstractive summarization. In: Proc. ACL; 2020. p. 1906–9.
17. Guu K, Lee K, Tung Z, Pasupat P, Chang M-W. REALM: Retrieval-augmented language model pretraining. In: Proc. ICML; 2020. p. 3929–38.
18. Karpukhin V, Oguz B, Min S, et al. Dense passage retrieval for open-domain question answering. In: Proc. EMNLP; 2020. p. 6769–81.
19. Izacard G, Grave E. Leveraging passage retrieval with generative models for open-domain QA. In: Proc. EACL; 2021. p. 874–80.
20. Borgeaud S, Mensch A, Hoffmann J, et al. Improving language models by retrieving from trillions of tokens. In: Proc. ICML; 2022. p. 2206–40.
21. Bodenreider O. The Unified Medical Language System (UMLS): integrating biomedical terminology. Nucleic Acids Res. 2004;32(Database issue):D267–70.
22. Donnelly K. SNOMED CT: The advanced terminology and coding system for eHealth. Stud Health Technol Inf. 2006;121:279–90.
23. Tsatsaronis G, Balikas G, Malakasiotis P, et al. An overview of the BIOASQ large-scale biomedical semantic indexing and question answering competition. BMC Bioinformatics. 2015;16:138.
24. Robertson S, Zaragoza H. The probabilistic relevance framework: BM25 and beyond. Found Trends Inf Retr. 2009;3(4):333–89.
25. Reimers N, Gurevych I. Sentence-BERT: Sentence embeddings using Siamese BERT networks. In: Proc. EMNLP-IJCNLP; 2019. p. 3982–92.
26. Ribeiro M, Singh S, Guestrin C. "Why should I trust you?" Explaining the predictions of any classifier. In: Proc. KDD; 2016. p. 1135–44.
27. Lundberg S, Lee S-I. A unified approach to interpreting model predictions. In: Proc. NeurIPS; 2017. p. 4765–74.
28. Zheng L, Chiang W-L, Sheng Y, et al. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. arXiv:2306.05685. 2023.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.